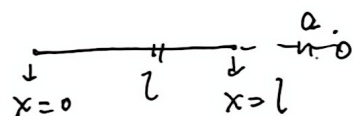


8.4



$x_0 = l + a$, 取 $dq = \lambda dx$.

$$r = l + a - x.$$

$$E = \int_0^l \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{(l+a-x)^2}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{l+a} \right)$$

$$= \frac{\lambda l}{4\pi\epsilon_0 a(l+a)}$$

若带正电, 则方向水平向右.

8.5 由对称性可知, 沿轴方向场强抵消.

又由左右对称, 沿直径方向分量抵消.

选取长 l , 宽 $Rd\theta = \frac{l d\theta}{2}$ 的长条.

$$\lambda' = \sigma l.$$

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{R} \sin\alpha, \quad \sin\alpha = \frac{\frac{l}{2}}{\sqrt{R^2 + (\frac{l}{2})^2}} = \frac{\sqrt{2}}{2}.$$

$$E = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} dE_1 \cos\theta = \frac{\sigma}{2\sqrt{2}\pi\epsilon_0} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\theta d\theta = \frac{\sigma}{\sqrt{2}\pi\epsilon_0}.$$

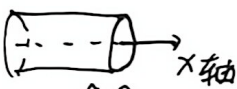
8.6 (1) $\lambda = \sigma dx$, $dE = \frac{\sigma dx}{2\pi\epsilon_0(a+x)}.$

$$E = \int_0^b dE = \frac{\sigma}{2\pi\epsilon_0} \ln\left(1 + \frac{b}{a}\right).$$

(2) 取 dx 宽长条, 它到中心线距离为 x .

$$dE_1 = \frac{\sigma dx}{2\pi\epsilon_0 \sqrt{x^2 + h^2}}, \quad dE_{1y} = dE_1 \cdot \frac{h}{\sqrt{x^2 + h^2}} = \frac{\sigma h dx}{2\pi\epsilon_0 (x^2 + h^2)}.$$

$$E = \int_{-\frac{b}{2}}^{\frac{b}{2}} dE_{1y} = \frac{\sigma}{\pi\epsilon_0} \arctan\left(\frac{b}{2h}\right)$$

8.7. 以柱线为x轴  取厚度为dx圆环, $dq = \frac{q}{l} dx$

由 $dE = \frac{1}{4\pi\epsilon_0} \frac{dq \cdot r}{(R^2 + r^2)^{\frac{3}{2}}}$
 r 为圆环到点距离.

设 $x = l + h$, $r = l + h - x$.

$$E = \int_0^l \frac{\frac{q}{l}}{4\pi\epsilon_0} \frac{(l+h-x)}{[R^2 + (l+h-x)^2]^{\frac{3}{2}}} dx$$

令 $u = l + h - x$
 $\Rightarrow \frac{q}{4\pi\epsilon_0 l} \int_h^{l+h} \frac{u du}{(R^2 + u^2)^{\frac{3}{2}}} = \frac{q}{4\pi\epsilon_0 l} \left(\frac{1}{\sqrt{R^2 + h^2}} - \frac{1}{\sqrt{R^2 + (l+h)^2}} \right)$

$R \rightarrow 0$ 时, $E = \frac{q}{4\pi\epsilon_0 l} \left(\frac{1}{h} - \frac{1}{l+h} \right)$

$\Rightarrow$ 圆柱极细时退化为线模型

8.18 $\oint_S E \cdot dS = \frac{\sum q_{in}}{\epsilon_0}$

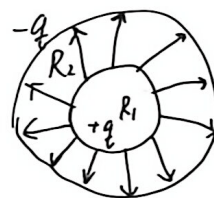
选半径为 r 的同心球面为高斯面.

$r < R_1$ 时, 内部无电荷, $\sum q_{in} = 0$, $E_1 = 0$

$R_1 < r < R_2$ 时, $\sum q_{in} = +q$, $E_2 \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$.

$$\Rightarrow E_2 = \frac{q}{4\pi\epsilon_0 r^2}$$

$r > R_2$ 时, $\sum q_{in} = q + (-q) = 0$, $E_3 = 0$



8.20 选长为 L , 半径为 r 的同轴圆柱面作为高斯面.
 高斯面侧面积为 $2\pi rL$.

$r < R_1$, $E_1 = 0$; $R_1 < r < R_2$, $q_{in} = \lambda_1 L$

$$E_2 \cdot 2\pi rL = \frac{\lambda_1 L}{\epsilon_0} \Rightarrow E_2 = \frac{\lambda_1}{2\pi\epsilon_0 r}$$

$r > R_2$ 时, $q_{in} = (\lambda_1 + \lambda_2)L$, $E_3 = \frac{\lambda_1 + \lambda_2}{2\pi\epsilon_0 r}$

8.22 (1) ① $r < R_1$ 时, $q_{in} = \rho_1 \pi r^2 L$
 $E \cdot 2\pi r L = \frac{\rho_1 \pi r^2 L}{\epsilon_0}$

$$E = \frac{\rho_1 r}{2\epsilon_0}$$

② $R_1 < r < R_2$ 时, $q_{in} = \rho_1 \pi R_1^2 L$

$$E = \frac{\rho_1 R_1^2}{2\epsilon_0 r}$$

③ $R_2 < r < R_3$ 时, $q_{in} = \rho_1 \pi R_1^2 L + \rho_2 \pi (r^2 - R_2^2) L$

$$E = \frac{\rho_1 R_1^2 + \rho_2 (r^2 - R_2^2)}{2\epsilon_0 r}$$

④ $r > R_3$ 时, $q_{in} = \rho_1 \pi R_1^2 L + \rho_2 \pi (R_3^2 - R_2^2) L$

$$E = \frac{\rho_1 R_1^2 + \rho_2 (R_3^2 - R_2^2)}{2\epsilon_0 r}$$

(2) $r > R_3$ 时, $E = 0 \Rightarrow \rho_1 R_1^2 + \rho_2 (R_3^2 - R_2^2) = 0$

$$\Rightarrow \frac{\rho_1}{\rho_2} = \frac{R_2^2 - R_3^2}{R_1^2}$$

此题下, 圆弧场强为

$$E = \frac{\lambda}{2\pi\epsilon_0 R} \sin \frac{\theta}{2}$$

对导线, $E_{\perp} = \frac{\lambda}{4\pi\epsilon_0 R}$, $E_{\parallel} = \frac{\lambda}{4\pi\epsilon_0 R}$, $E_2 = \frac{\lambda \cos \frac{\theta}{2}}{2\pi\epsilon_0 R}$

$$E_2 = 2(E_{\perp} \cos \frac{\theta}{2} - E_{\parallel} \sin \frac{\theta}{2}) = \frac{\lambda \cos \frac{\theta}{2}}{2\pi\epsilon_0 R}, \text{ 水平向左. 故 } E_0 = 0 \Rightarrow E = E_2 \Rightarrow \sin \frac{\theta}{2} = \cos \frac{\theta}{2}$$

$$\Rightarrow \theta = \frac{\pi}{2} \text{ 即可}$$

8.23 $\rho(r) = \frac{\rho_0}{[1 + (\frac{r}{a})^2]^2}$

选长为 L , 半径为 r 的高斯面.

$$q_{in} = \int_0^r \rho(r') \cdot 2\pi r' L dr' = 2\pi L \rho_0 \int_0^r \frac{r' dr'}{[1 + (\frac{r'}{a})^2]^2}$$

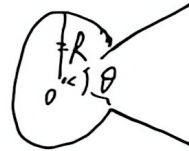
令 $u = 1 + (\frac{r'}{a})^2$, $du = \frac{2r'}{a^2} dr'$, $\Rightarrow r' dr' = \frac{a^2}{2} du$

$$q_{in} = 2\pi L \rho_0 \cdot \frac{a^2}{2} \int_1^{1+(\frac{r}{a})^2} \frac{du}{u^2} = \pi L \rho_0 a^2 \left(1 - \frac{1}{1+(\frac{r}{a})^2}\right)$$

由高斯定理: $E \cdot 2\pi r L = \frac{q_{in}}{\epsilon_0}$

$$\Rightarrow E = \frac{\rho_0 r}{2\epsilon_0 [1 + (\frac{r}{a})^2]}$$

补充:



由对称性, 圆弧在 O 点产生的场强向右.

对圆弧 (R, λ) , 圆心角为 $\alpha = 2\pi - \theta$, 将其放置坐标系中.

$$dl = R d\theta, dq = \lambda R d\theta$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{R^2} = \frac{\lambda d\theta}{4\pi\epsilon_0 R}, dE_y = dE \cos \theta$$

$$E = \int_{-\frac{\alpha}{2}}^{\frac{\alpha}{2}} \cos \theta \cdot \frac{\lambda}{4\pi\epsilon_0 R} d\theta = \frac{2\lambda}{4\pi\epsilon_0 R} \cdot 2\sin \frac{\alpha}{2}$$